

Deep learning (DL) is a type of ML algorithm that uses neural networks (NN), a modelling approach inspired by how our brains work. It comprises millions of neurons, connected to each other, organised in hundreds or more layers.

To address complex cyber attacks and threats, several organisations are implementing AI and ML based security solutions and technologies.

AI and ML have evolved as an optimal solution in cyber security, with solution techniques/algorithms being applied across all market sectors. They play a crucial role in the development of automated security systems, natural language processing, face recognition, and autonomous threat detection.

Popular AI/ML approaches to cyber security

Some of the most prominent and popular ways in which AI and ML detect cyber threats are described below briefly.

Malware detection and identification: In this approach, AI and ML algorithms help in identifying malicious files and filter them before they reach the end user. Many different

AI and ML approaches have been used to detect malware. These solutions can detect, respond, and remediate in real-time. Most prominent among these are:

- Machine learning and data mining to look for malware source code repositories using a technique called ‘SourceFinder’ and analyse them based on characteristics and properties
- Machine learning to look for a particular string within files that could indicate the presence of malware or malicious code and classify them
- Usage of AI/ML to detect patterns in binary executable files and determine if they are malicious
- Utilising visual binary patterns identified in the code and a type of self-organising network that adapts over time

Behavioural threat analysis:

Threat analysis is a cyber security strategy that aims to assess an organisation’s security protocols, processes and procedures to identify threats, vulnerabilities, and gather knowledge of a potential attack in advance. A threat analysis consists of the information and assets that

need to be protected in terms of confidentiality, integrity, and availability. Some of the techniques used to identify the threats are:

- ML techniques called user and event behavioural analytics (UEBA) to analyse and recognise typical behaviours and patterns in user accounts and endpoints. These can detect security incidents that violate predefined operational rules, employ novel attack patterns, or span multiple organisational systems and data sources.
- Semi-supervised learning is another approach used for threat detection.

Spam detection: Spam emails may have inappropriate contents, links and attachments, which can lead to security issues. Uninvited bulk emails generally belong to the category of spam. AI and ML models can be used to detect spam by analysing the content of the mail/message and looking for patterns.

AI is able to detect that messages are spam without requiring any human intervention. Some of the spam detection algorithms are:

- Bayes algorithm that helps to filter out some spam emails
- AI and ML based spam detection unsupervised text mining model used to detect the possibility of false reviews

Network intrusion detection:

AI and ML based intrusion detection systems (IDS) develop intelligent systems to detect, classify, and respond to cyber attacks. IDSs identify malicious behaviour and stop it before it causes any damage.

An IDS is usually configured with a set of rules such as use of certain words in the subject line of an email message or sending of too many messages in a given time period. It helps to generate alerts when it detects an event related to an attack or intrusion attempt.

IDS can be implemented as a standalone system or as an add-on module to security software — for example, antivirus programs.

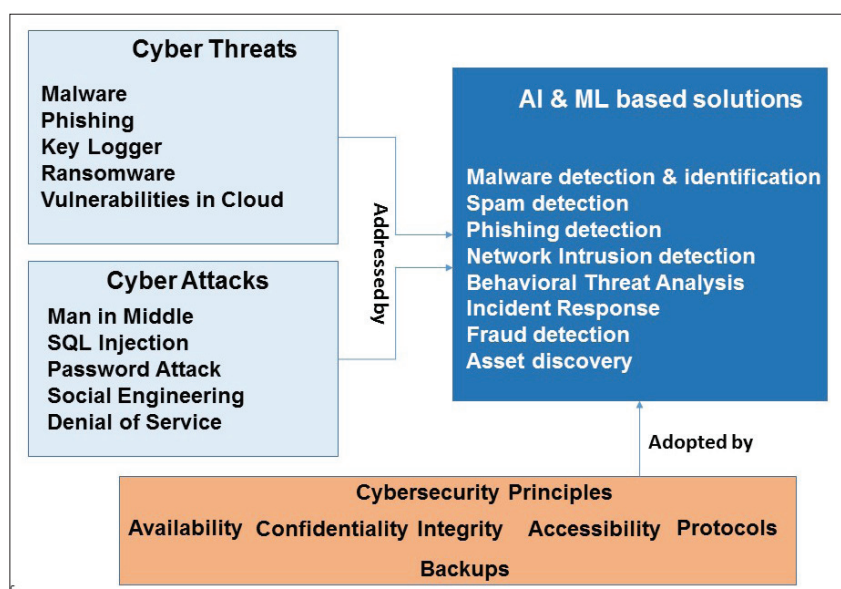


Figure 2: AI and ML based cyber security solutions